

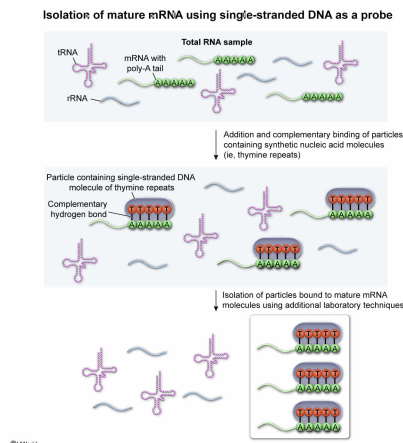
In an experiment, a researcher wants to isolate mature mRNA from a sample of total RNA using specialized particles containing long single-stranded nucleic acid molecules (either DNA or RNA) of a synthetic nature. Which of the following single-stranded molecules would most likely be used as a probe to isolate the mature mRNA?

- ☐ A. A DNA molecule consisting of uracil repeats [14%]
- ✓ ☒ B. A DNA molecule consisting of thymine repeats [41%]
- ☐ C. An mRNA molecule consisting of adenine repeats [33%]
- ☐ D. An mRNA molecule consisting of cytosine repeats [11%]

Correct

42% answered correctly

Explanation:



In **eukaryotes**, **genes** are initially **transcribed** into pre-messenger RNA (pre-mRNA) that is then **modified** to form mature mRNA (ie, the mRNA that is **translated** into proteins). Modifications to pre-mRNA include **splicing** (ie, removal) of noncoding introns as well as the addition of a **3' poly-adenine (poly-A) tail** and a **5' GTP cap**. Specifically, the poly-A tail and GTP cap allow mature mRNA to exit the nucleus as well as help prevent mRNA from being degraded in the cytoplasm.

In this scenario, a researcher wants to **isolate mature mRNA** from a sample of total RNA using specialized particles that contain long single-stranded nucleic acid molecules of a synthetic nature (either synthetic DNA or RNA). In addition to mature mRNA, the total RNA sample would include other RNA types such as **tRNAs** (add amino acids to the growing polypeptide during translation at the ribosome) and **rRNAs** (components of ribosomes). Therefore, the researcher should use a nucleic acid molecule (ie, probe) that can bind to a structure common among **all mature mRNAs**.

Because a **3' poly-A tail** is added to all mature mRNA molecules, a **single-stranded DNA molecule** consisting of **thymine repeats** could be used as a probe because these thymine nucleotides can form **complementary** hydrogen bonds (ie, **Watson-Crick base pairs**) with the adenine nucleotides of the mature mRNA poly-A tail. This would allow the mature mRNA to bind the particles and consequently be isolated from the total RNA sample.

(Choice A) Although uracil repeats could also form complementary bonds with adenine nucleotides comprising the poly-A tail of mature mRNA, uracil is found **only in RNA** (*not* DNA) molecules.

(Choices C and D) Adenine would form complementary bonds with either *thymine* or *uracil*, and cytosine would form complementary bonds with *guanine*. Therefore, these nucleotides would *not* form complementary bonds with the *adenine* nucleotides of the poly-A tail in mature mRNA molecules and would *not* be useful as probes in this experiment.

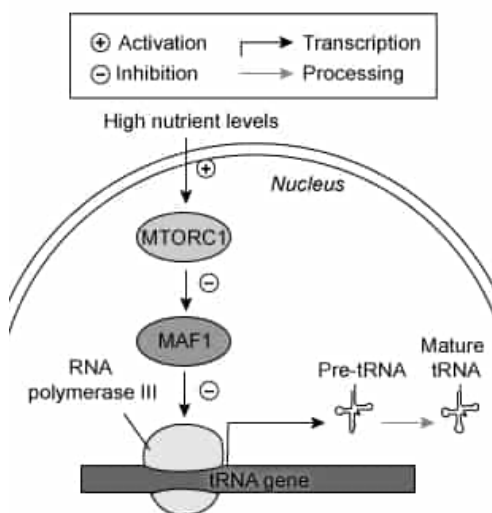
Educational objective:

In eukaryotes, genes are initially transcribed into pre-mRNA molecules that are modified to form mature mRNA molecules. These modifications include splicing of introns and adding both a 3' poly-A tail (ie, multiple adenine nucleotides) and 5' GTP cap. The poly-A tail and GTP cap protect the mRNA from being degraded in the cytoplasm.

Time spent: 0 sec
Subject:
Biology

QID: 403127
System:
Molecular Biology

The diagram below shows a model of how nutrient availability influences transfer RNA (tRNA) production in human cells. When activated, the MAF1 protein inhibits RNA polymerase III, and low nutrient levels inhibit the mTORC1 protein in this pathway.



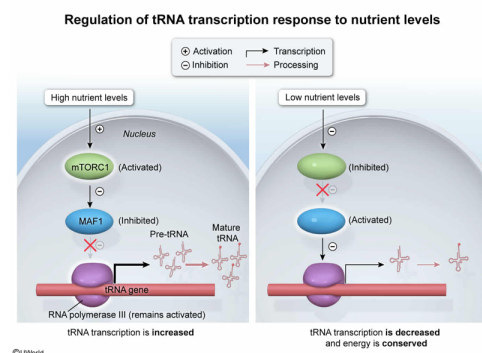
Given this information, which of the following statements is most likely true?

- ☐ A. MAF1 decreases tRNA transcription when nutrient levels are high to increase the storage of amino acids. [30%]
- ☐ B. mTORC1 increases tRNA transcription when nutrient levels are low to increase protein availability. [18%]
- ☐ C. mTORC1 increases tRNA transcription when nutrient levels are high to increase MAF1 production and activity. [8%]
- ✓ ☒ D. MAF1 decreases tRNA transcription when nutrient levels are low to promote energy conservation. [41%]

Correct

42% answered correctly

Explanation:



During **transcription**, which requires cellular energy (eg, ATP), the nucleotide sequence of a DNA region is used as a template by RNA polymerase to synthesize an RNA strand. Several **RNA types** are transcribed, including **protein-coding** messenger RNA (mRNA) and **noncoding** **ribosomal RNA (rRNA)**, **microRNA (miRNA)**, **small nuclear RNA (snRNA)**, **small interfering RNA (siRNA)**, and **transfer RNA (tRNA)**.

In **prokaryotes**, a **single** type of RNA polymerase is used for both coding and noncoding RNA transcription. However, in **eukaryotes** three RNA polymerase types are found:

- **RNA polymerase I** carries out transcription of rRNA

- **RNA polymerase II** carries out transcription of mRNA, some miRNAs, and some snRNAs
- **RNA polymerase III** carries out transcription of tRNAs as well as some rRNAs, miRNAs, and snRNAs

The diagram shows how nutrient availability in human cells influences production of tRNAs, which are first transcribed into pre-tRNA molecules and then processed to form mature tRNAs. To facilitate protein **translation**, tRNAs "charged" (ie, bound) with specific amino acids have **anticodons** that bind specific mRNA codons. This allows tRNAs to add corresponding **amino acids** to growing polypeptides during translation.

In the diagram, *high* nutrient levels *activate* mTORC1, which *inhibits* MAF1. The question states that when MAF1 is activated, it *inhibits* RNA polymerase III activity and therefore tRNA transcription. Accordingly, MAF1 *inhibition* by mTORC1 when nutrient levels are high *prevents* MAF1 from inhibiting RNA polymerase III, *increasing* (not decreasing) tRNA transcription (**Choice A**).

The question also states that *low* nutrient levels *inhibit* mTORC1, which would likely prevent it from inhibiting MAF1. Therefore, in **low nutrient conditions**, MAF1 would likely be *activated* and **inhibit RNA polymerase III**, leading to **decreased** (not increased) **tRNA transcription (Choice B)**. By *decreasing* energy use for tRNA production, **energy conservation** would likely be **promoted** in the cell.

(Choice C) In *high* nutrient conditions, *inhibition* of MAF1 by mTORC1 would *increase* RNA polymerase III activity and tRNA transcription. However, mTORC1 *inhibits* MAF1 and would likely *not* promote increased MAF1 activity.

Educational objective:

During transcription, RNA polymerase transcribes RNA molecules using a DNA template. In eukaryotes, different RNA polymerase types are responsible for transcribing different RNA types. For example, transcription of transfer RNA (tRNA) is carried out by RNA polymerase III.

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